

5.3.2

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Question

The pair of linear equations

$$\frac{3x}{2} + \frac{5y}{3} = 7 \quad \text{and} \quad 9x + 10y = 14$$

is

- ☐ consistent
- ☐ inconsistent
- ☐ consistent with one solution
- ☐ consistent with many solutions

Step 1: Clear Fractions

Multiply the first equation by 6:

$$9x + 10y = 42 \quad (1)$$

The second equation is

$$9x + 10y = 14 \quad (2)$$

Step 2: Augmented Matrix

The system can be written as:

$$\left(\begin{array}{cc|c} 9 & 10 & 42 \\ 9 & 10 & 14 \end{array} \right) \quad (3)$$

Subtracting R_1 from R_2 :

$$\left(\begin{array}{cc|c} 9 & 10 & 42 \\ 0 & 0 & -28 \end{array} \right) \quad (4)$$

This implies $0x + 0y = -28$, which is a contradiction.

Step 3: Conclusion

Therefore, the two equations represent **parallel distinct lines**.

The system is inconsistent.

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Define x-range
x = np.linspace(-5, 10, 400)

# Define the two lines:  $y = (\text{constant} - 9x)/10$ 
y1 = (42 - 9*x) / 10    # from  $9x + 10y = 42$ 
y2 = (14 - 9*x) / 10    # from  $9x + 10y = 14$ 

# Plotting
plt.figure(figsize=(6,6))
plt.plot(x, y1, label=r'$9x+10y=42$', color='blue')
plt.plot(x, y2, label=r'$9x+10y=14$', color='red')
plt.xlabel('x')
plt.ylabel('y')
plt.axhline(0, color='black', linewidth=0.5) # x-axis
plt.axvline(0, color='black', linewidth=0.5) # y-axis
plt.title('Lines:  $9x+10y=42$  and  $9x+10y=14$ ')
```

C Code

```
#include <stdio.h>

int main() {
    double a1=9, b1=10, c1=42;
    double a2=9, b2=10, c2=14;

    double r1 = a1/a2;
    double r2 = b1/b2;
    double r3 = c1/c2;

    if(r1 == r2 && r2 == r3)
        printf("Consistent: Infinite solutions\n");
    else if(r1 == r2 && r2 != r3)
        printf("Inconsistent system (No solution)\n");
    else
        printf("Consistent: Unique solution\n");

    return 0;
}
```

Plot

beamer/figs/fig1.png

Figure: Graph of the lines $9x + 10y = 42$ and $9x + 10y = 14$.